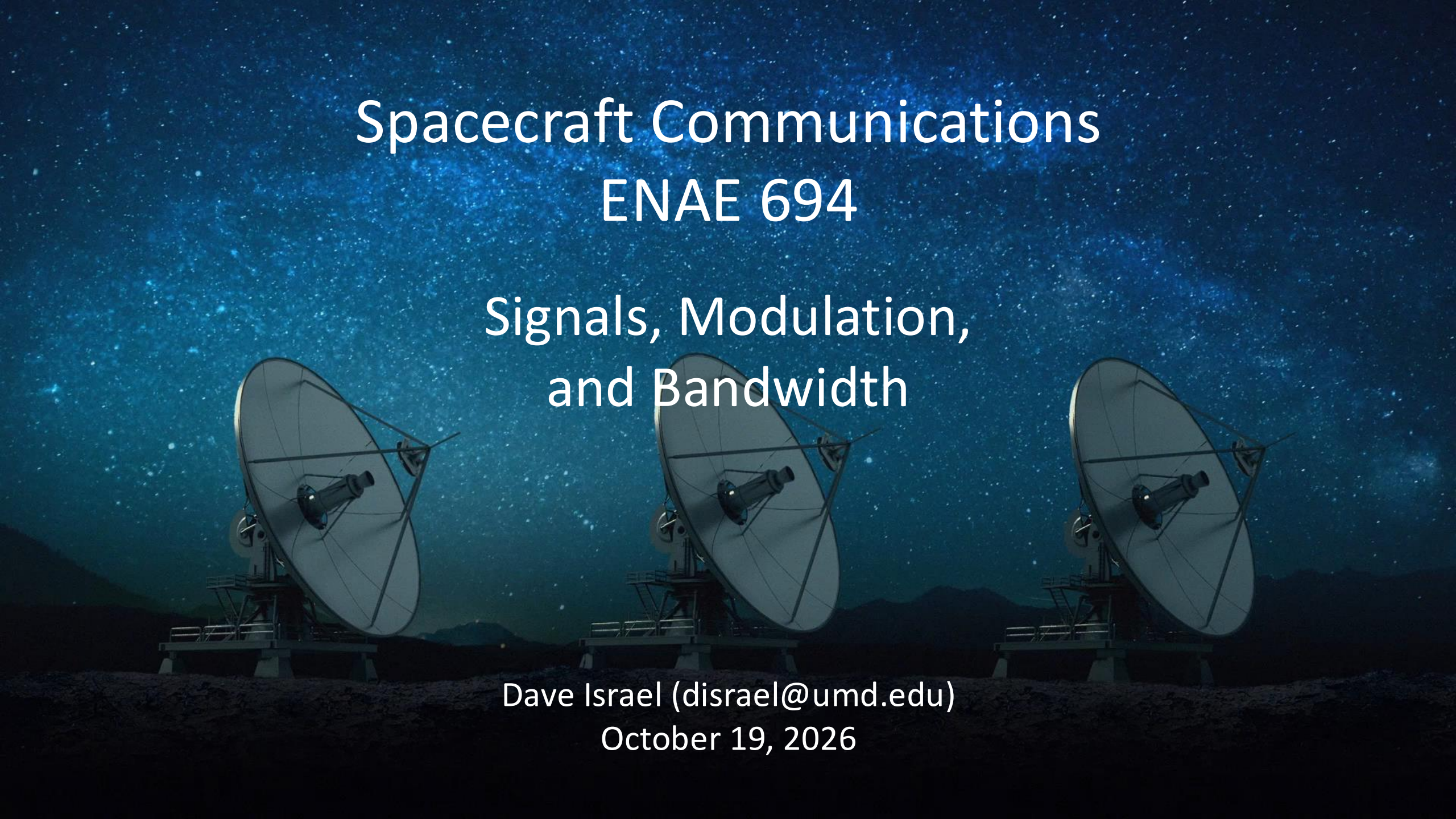


Spacecraft Communications
ENAE 694

Signals, Modulation, and Bandwidth

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October 19, 2026

Tonight's Agenda

- Self-Contained Homework 3 Review
- Part 1: Carriers, Bits, Symbols, and Digital Modulation
- Part 2: Pulse Shaping, Bandwidth, and Link Performance

Homework 3, Problem 1: The Question

A circular reflector has D = 0.80 m, f = 7.50 GHz, and aperture efficiency 0.55.

(a) Find peak gain in dBi and full half-power beamwidth (HPBW) in degrees.

(b) Double D at fixed f. Find gain change, new gain, and new HPBW.

(c) Restore D and double f to 15.00 GHz. Find gain change and HPBW. Why is this insufficient to determine the received-power change?

Homework 3: Gain and Beamwidth Worked Answer

Case	Gain [G] (dBi)	Change (dB)	HPBW (°)
D = 0.80 m, f = 7.50 GHz	33.37	Reference	3.50
Double D to 1.60 m	39.39	+6.02	1.75
Double f to 15.00 GHz	39.39	+6.02	1.75

$$[G] = 17.81 + 20 \log_{10}(D) + 20 \log_{10}(f)$$

HPBW ≈ 21/(fD)°. D in m, f in GHz; efficiency 0.55 included.

Homework 3: Why Gain Alone Is Insufficient

Doubling f changes both antenna gain and free-space loss.

$$\Delta[S] = \Delta[G_{TX}] + \Delta[G_{RX}] - \Delta[L_p] - \Delta[L_{other}]$$

What Stays Fixed?	Changes at Twice the Frequency	Received Power
Ground antenna gain	+6.0 + 0.0 − 6.0 dB	0.0 dB change
Both physical apertures	+6.0 + 6.0 − 6.0 dB	+6.0 dB change

We must know the ground-antenna assumption and any other frequency-dependent losses.

Homework 3, Problem 2: The Question

Aligned, polarization-matched [P_{r0}] = −118.00 dBW. Required received power = −123.50 dBW. All other losses are included.

Transmit HPBW = 3.50°, error = 0.875°. Receive HPBW = 1.40°, error = 0.350°. Both polarizations are linear.

(a) Find each pointing loss. (b) At 45° polarization misalignment, find polarization loss, actual received power, and margin. Does it meet the threshold?

(c) Repeat power and margin for 60°. Explain how polarization realignment can restore the link without changing peak antenna gain.

Homework 3: Pointing Loss
Worked Answer

$[L_{point}] = 12(\theta / HPBW)^2$, using full HPBW and consistent angular units.

Transmit: $[L_{pt}] = 12(0.875 / 3.50)^2 = 0.75$ dB

Receive: $[L_{pr}] = 12(0.350 / 1.40)^2 = 0.75$ dB

Total pointing loss = 0.75 + 0.75 = 1.50 dB. Subtract it once.

The supplied near-boresight approximation does not describe sidelobes or nulls.

Homework 3: Polarization and Margin Worked Answer

Axes Differ By	$[L_{pol}]$ (dB)	$[P_r]$ (dBW)	Margin (dB)
45°	3.01	−122.51	+0.99: meets threshold
60°	6.02	−125.52	−2.02: fails threshold

$[L_{pol}] = -10 \log_{10}(\cos^2\psi)$
$[P_r] = -118.00 - 1.50 - [L_{pol}]$
Margin = $[P_r]$ − (−123.50 dBW)

Homework 3, Problem 3: The Question

Nadir-pointed science + continuous communications through a 12-minute overhead pass.

Time (min)	Fixed Gain Offset	Extra Path Loss
0–2 / 10–12	−8 dB	4 dB
2–5 / 7–10	−2 dB	1 dB
5–7	0 dB	0 dB

At midpass, peak margin = +5 dB. Margin ≥ 0 is usable.

Find fixed and ideal gimbaled margins. Which meets both requirements?
The gimbal holds peak gain, with no added loss. Why can’t we rotate the spacecraft?

Homework 3: Margin During the Pass Worked Answer

Time (min)	Fixed Gain (dB)	Extra [L_p] (dB)	Fixed Margin (dB)	Gimbal Margin (dB)
0–2	−8	4	−7	+1
2–5	−2	1	+2	+4
5–7	0	0	+5	+5
7–10	−2	1	+2	+4
10–12	−8	4	−7	+1

Margin = 5 + gain relative to peak − extra path loss.
Fixed: usable only 2–10 min. Gimbal: usable throughout the pass.
The gimbal restores peak gain; it does not remove the extra path loss.

Homework 3, Problem 4: The Question

Keep the science and continuous-communications requirements from Problem 3. Installed antenna-system mass limit = 5.0 kg.

Option A: 2.0 kg fixed antenna. Option B: 6.0 kg gimbaled antenna. Both have the RF performance used in Problem 3. Mass includes feed, supports, and mechanisms.

(a) Which requirements does each option meet? Does either meet all? (b) Recommend a feasible change and state its cost or performance consequence.

(c) Why does antenna mass location affect spacecraft slewing and pointing even when total mass is unchanged?

Homework 3: Mass and Pointing Worked Answer

Option	Installed Mass	Communications + Science	All Requirements?
A: Fixed	2.0 kg; 3.0 kg below limit	Cannot maintain both through full pass	No: loses link at pass ends
B: Gimbaled	6.0 kg; 1.0 kg above limit	Tracks station while science stays nadir-pointed	No: exceeds mass limit

Neither option meets every requirement as proposed.

The limit is 5.0 kg. Turning the spacecraft toward the station would violate the required nadir science attitude.

Homework 3: The Engineering Trade

Reallocate at least 1.0 kg to permit the 6.0 kg gimbaled system. Other hardware must give up that mass allocation.

Alternatively, redesign the gimbaled system to 5.0 kg or less while retaining RF and pointing performance. Verify the redesigned hardware.

Mass farther from a rotation axis contributes more rotational inertia: $I = \Sigma mr^2$ for point masses. This affects slew torque and settling.

*Problem 4(c) is outside this course's scope. Everyone receives full credit for that part.

Clarifications From October 5

An absorbing atmosphere weakens the signal and emits thermal noise.

Off the lunar disk means the line of sight is outside the apparent Moon.

S/N_0 compares signal power with noise per hertz. Its units are dB-Hz.

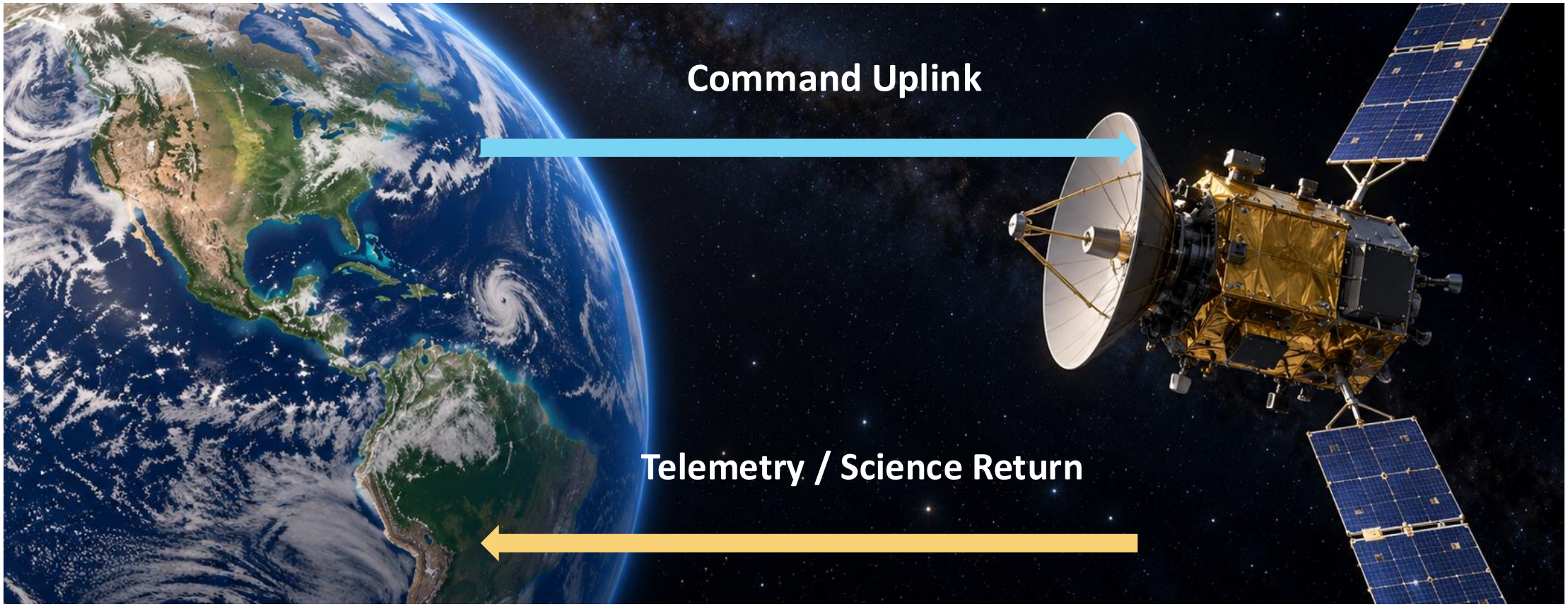
Noise factor F is linear. Noise figure NF = $10 \log_{10}(F)$ is in dB.

T_0 = 290 K: a common reference near room temperature.

It makes component NF values comparable. It is not the actual antenna or hardware temperature.

Another reference changes reported NF; consistent $T_e = (F - 1)T_0$ gives the same physical noise.

What Information Does a Signal Carry?

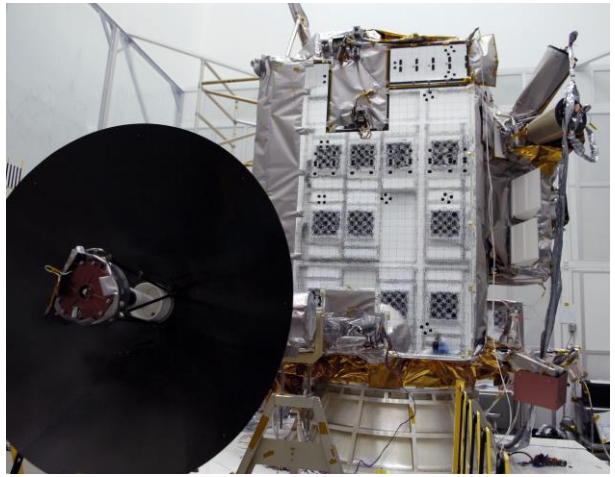


The waveform also provides timing, range, and Doppler measurements.

*Conceptual AI-generated illustration [15]. Sizes, distance, and signal paths are schematic.

The Spacecraft Communications Chain

SPACECRAFT



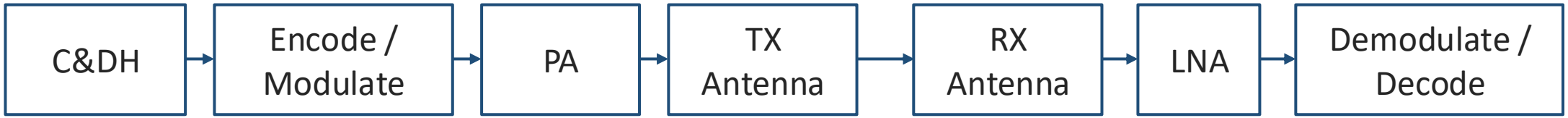
Avionics and RF hardware inside the spacecraft bus

Antenna and feed launch the radio wave

GROUND STATION



Return Link

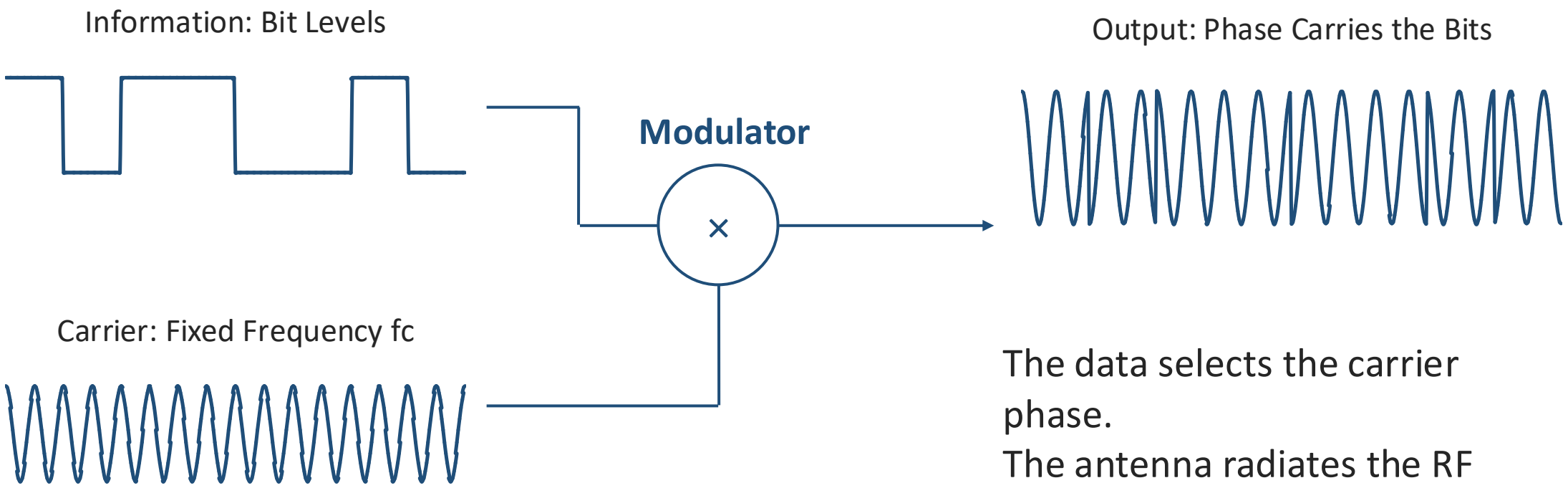


C&DH collects data. PA = power amplifier.
Low-loss RF feed connects PA to antenna.

Ground receiver and mission-data processing.
Keep loss before the LNA small.

Photos: LRO, NASA/GSFC [11]; Goldstone, NASA/JPL-Caltech [12]. Functional locations are schematic; DSN beam-waveguide optics carry RF to receiver equipment.

Why Put Information on a Carrier?

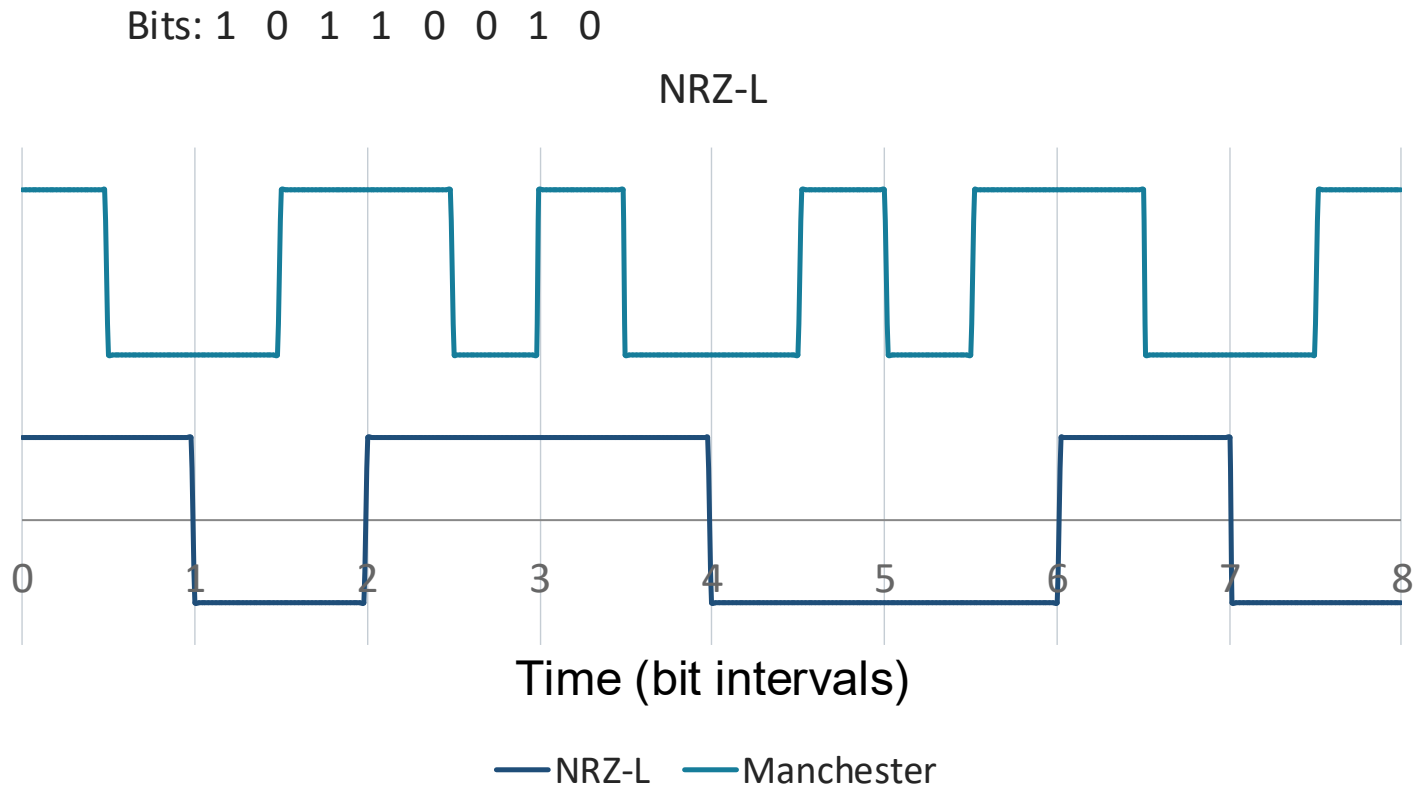


The data selects the carrier phase.
The antenna radiates the RF output.

Low-rate information moves to a band suited to antennas and spectrum allocation.

Illustrative BPSK. Only a few RF cycles per bit are drawn so the phase changes are visible. f_c is the carrier frequency.

Bits, Line Coding, and Symbols



A bit is a binary value. A symbol is one selected transmitted waveform.

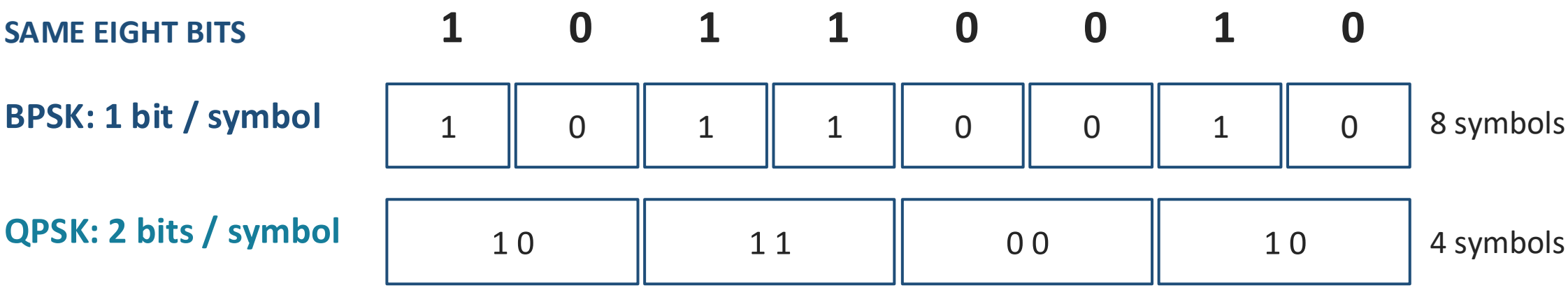
Line coding turns a bit stream into a baseband waveform.

NRZ and Manchester encode the same bits with different transitions.

Bit Rate and Symbol Rate

Bit rate R_b: information bits per second.

Symbol rate R_s: selected waveforms per second.

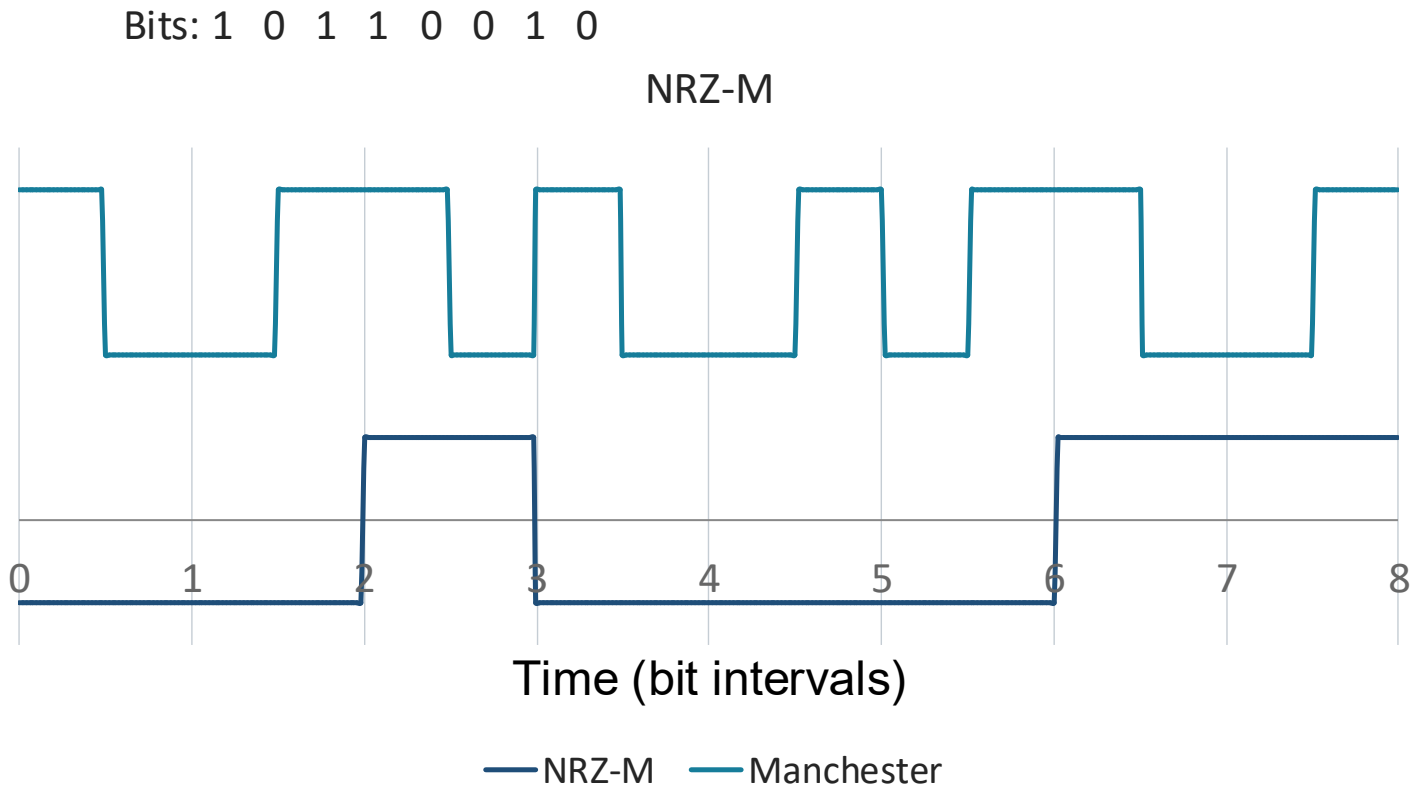


$R_s = R_b$ / (bits per symbol)

At 1.0 Mbit/s: BPSK = 1.0 Msymbol/s; QPSK = 0.5 Msymbol/s.

Two identical consecutive states still occupy two symbol intervals.

Line Coding Changes the Transitions

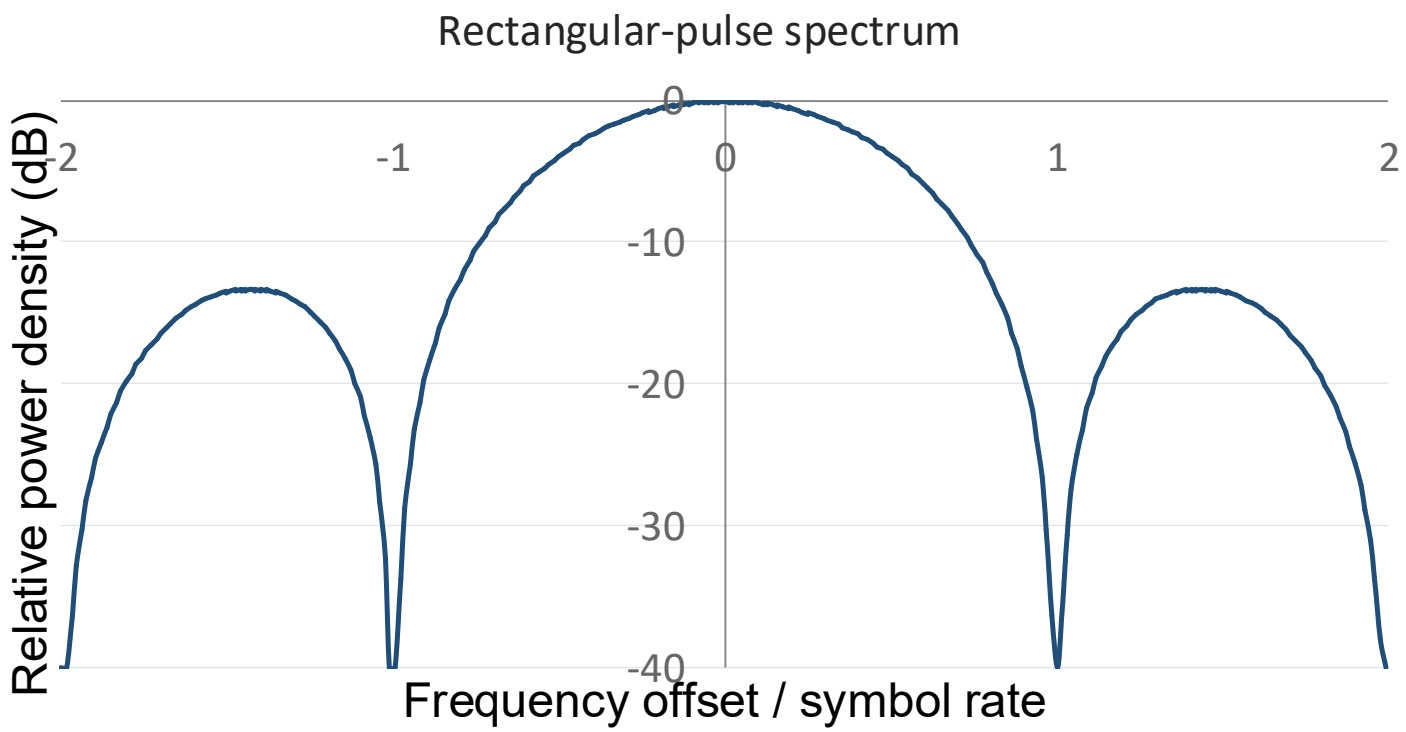


NRZ-L holds a level for each bit interval.

NRZ-M changes level on a chosen bit value. Manchester changes at every bit center.

More transitions can aid timing recovery, but require more bandwidth.

Time Domain and Frequency Domain

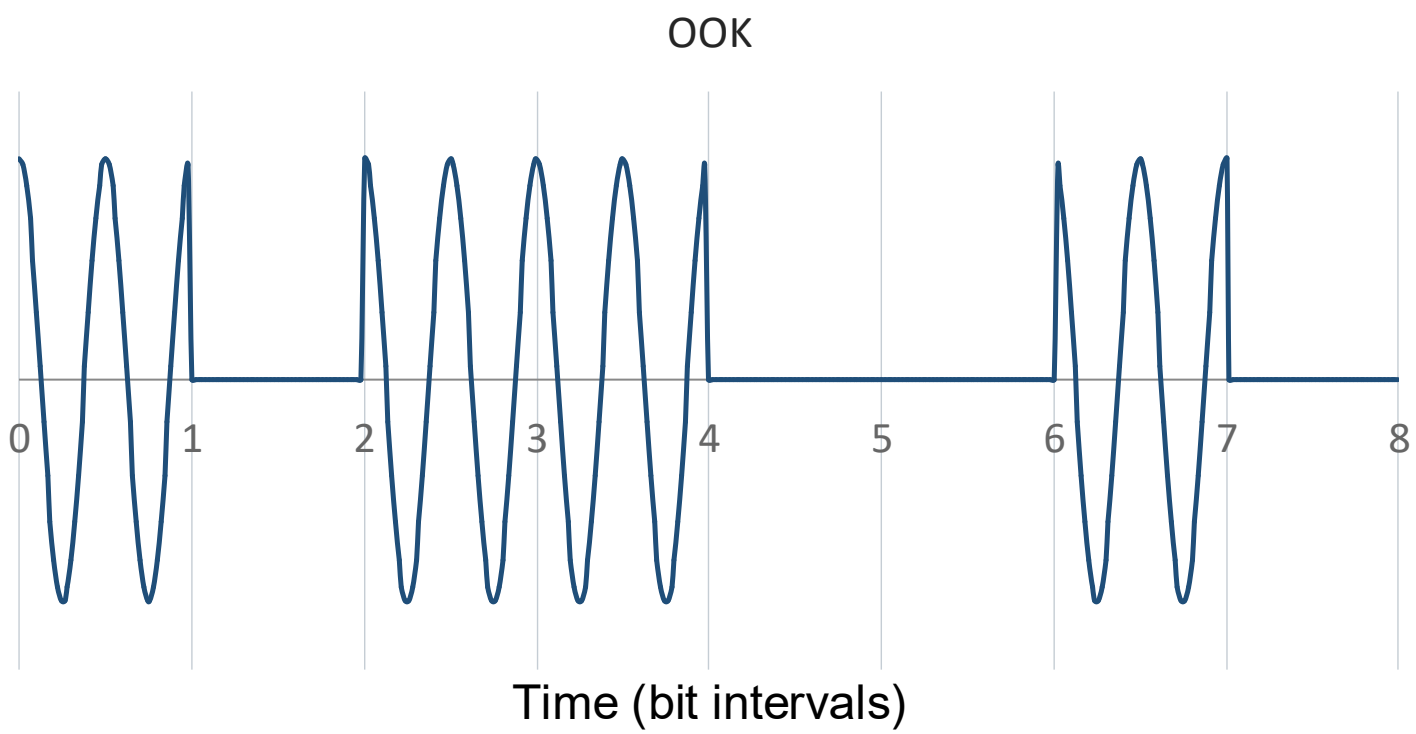


A waveform shows how the signal changes with time.

The Fourier transform gives its frequency-domain description.

Abrupt transitions spread signal energy into sidelobes.

On-Off Keying

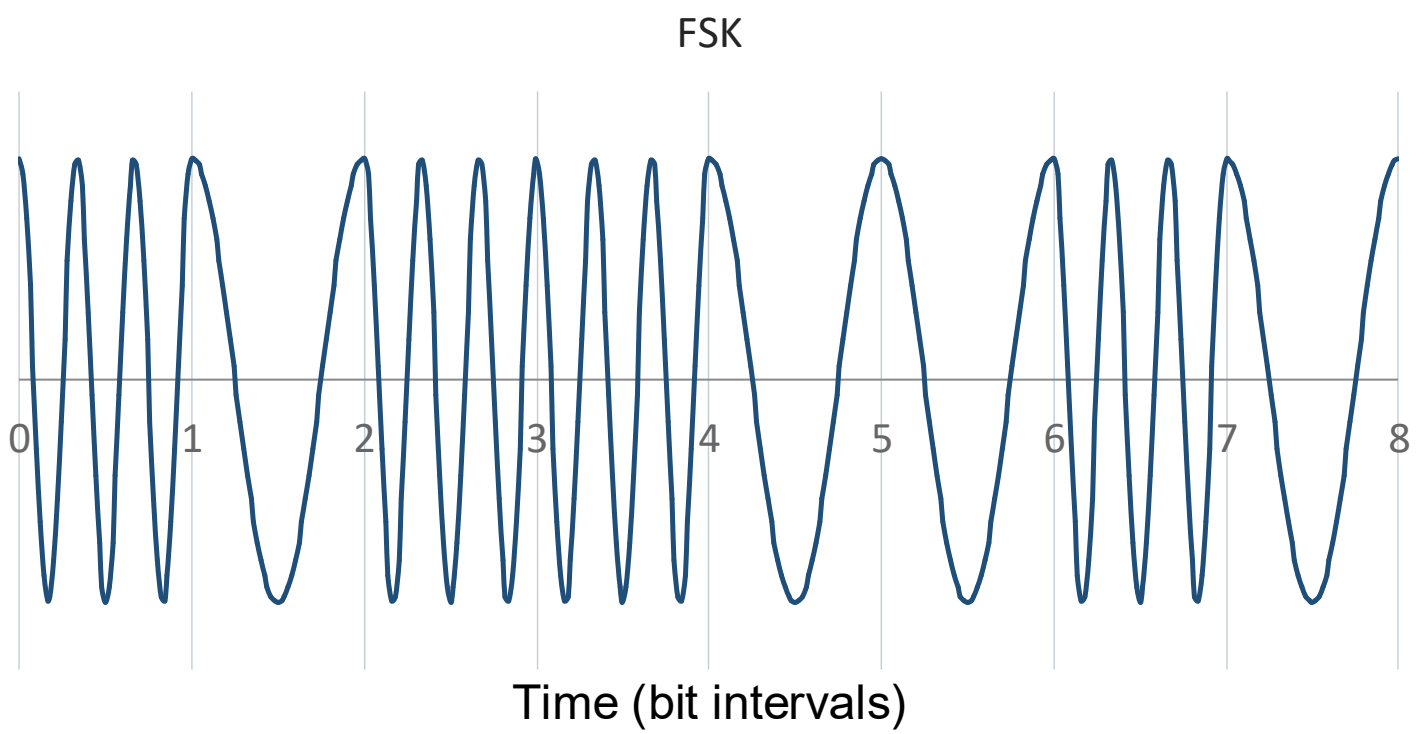


One state sends a carrier. The other sends no carrier.

The information changes the carrier amplitude.

For unbalanced on/off levels, a residual carrier component may remain.

Frequency-Shift Keying

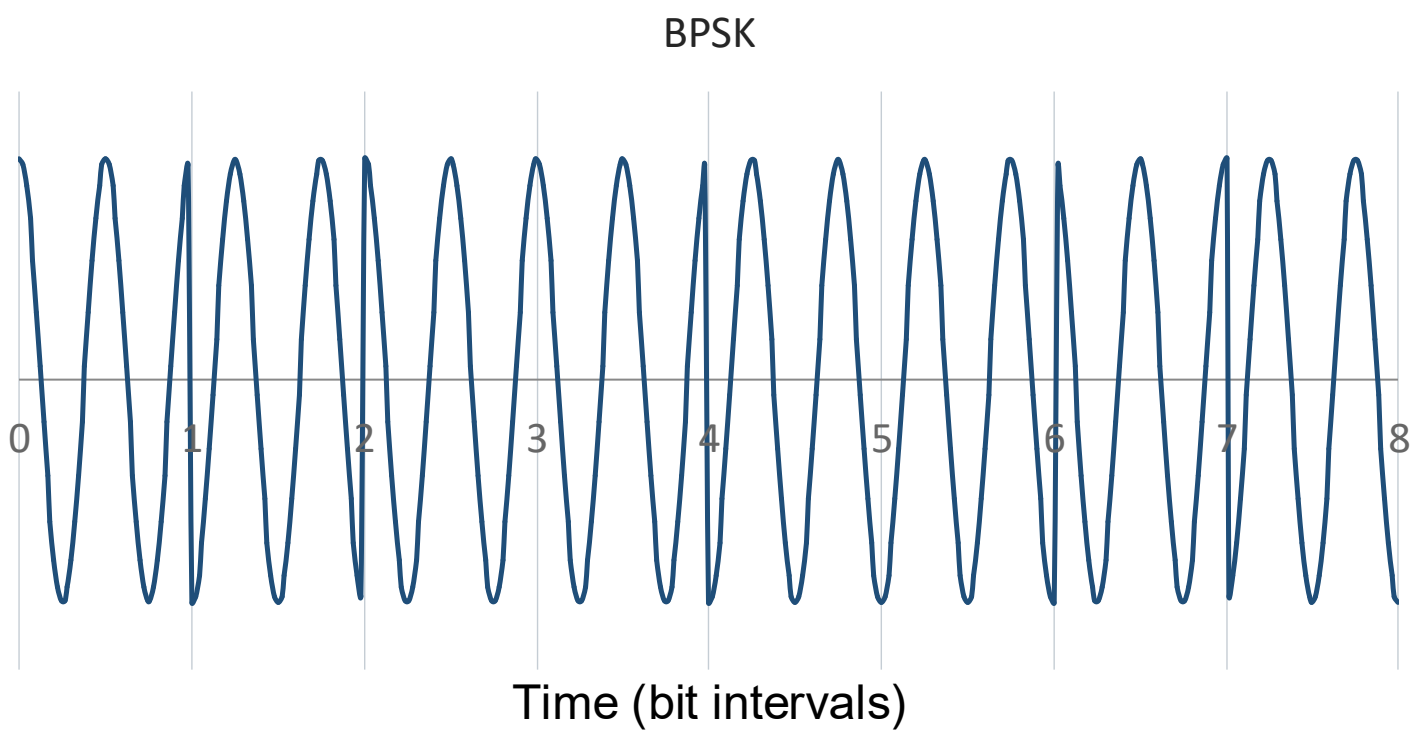


A 0 and a 1 select different carrier frequencies.

The receiver decides which tone is present.

Required bandwidth depends on tone spacing and pulse shaping.

BPSK: A Bit Selects the Carrier Phase

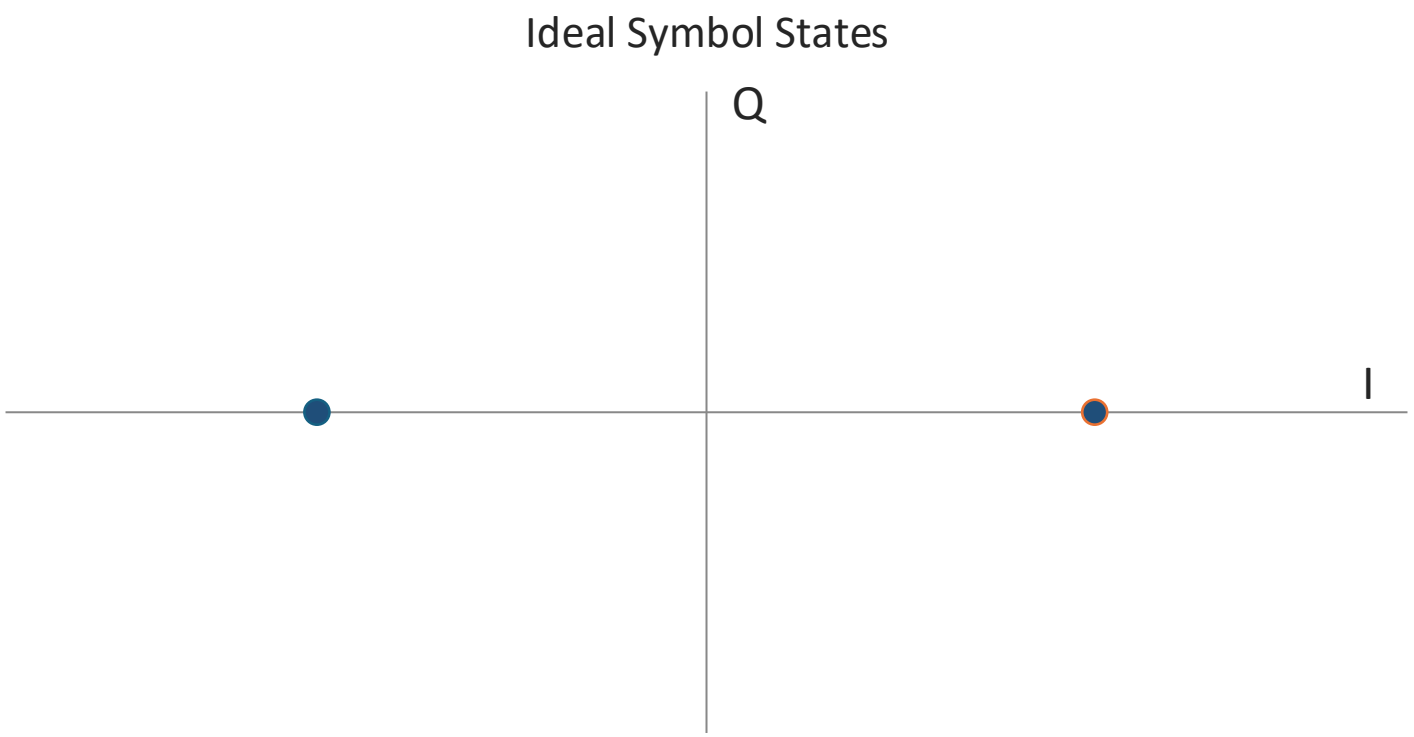


Binary phase-shift keying uses two phases separated by 180°.

One bit per modulation symbol.

For rectangular bit pulses, null-to-null RF bandwidth is approximately $2R_b$.

BPSK: Two Possible Decisions

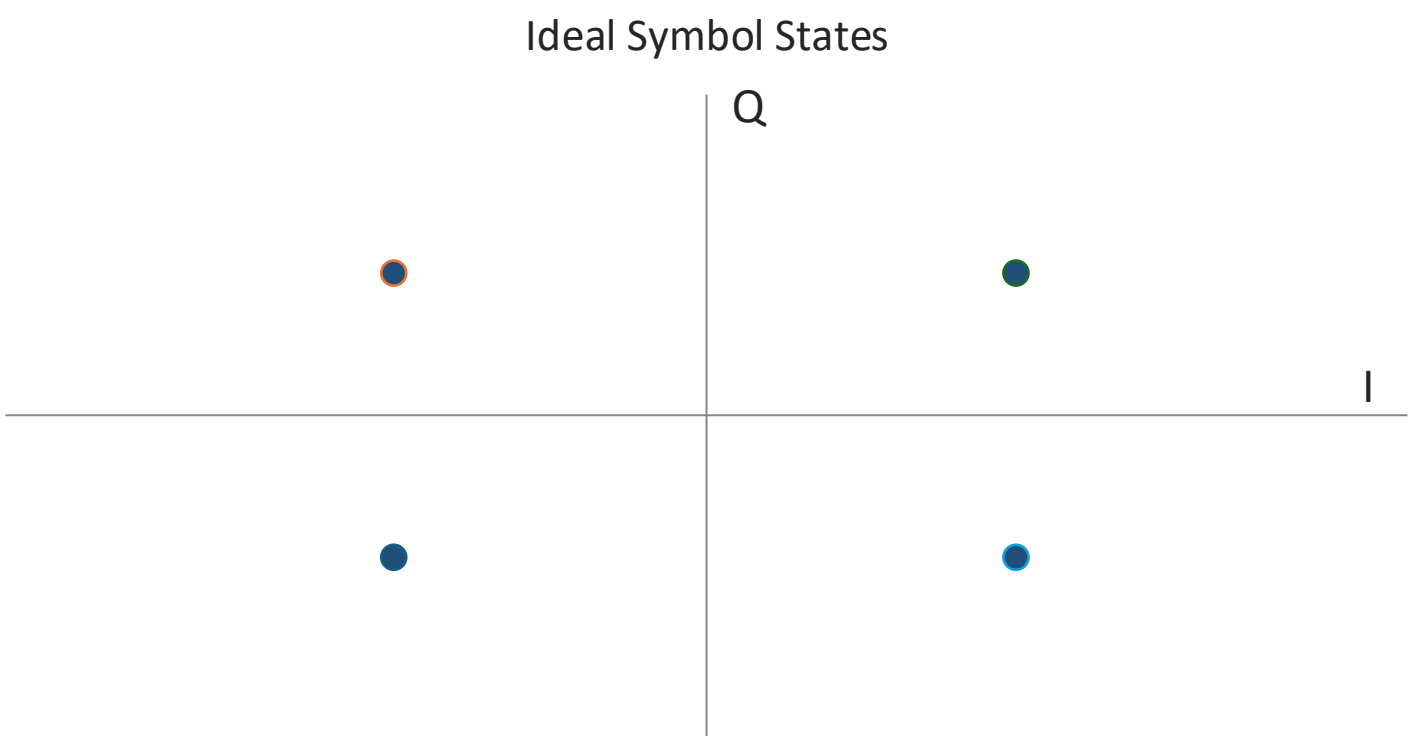


A constellation shows symbol states after carrier removal.

I = in-phase (horizontal). Q = quadrature (vertical), relative to the carrier reference.

Ideal BPSK has two I values and Q = 0. Each point represents one possible symbol.

QPSK: Two Bits per Symbol

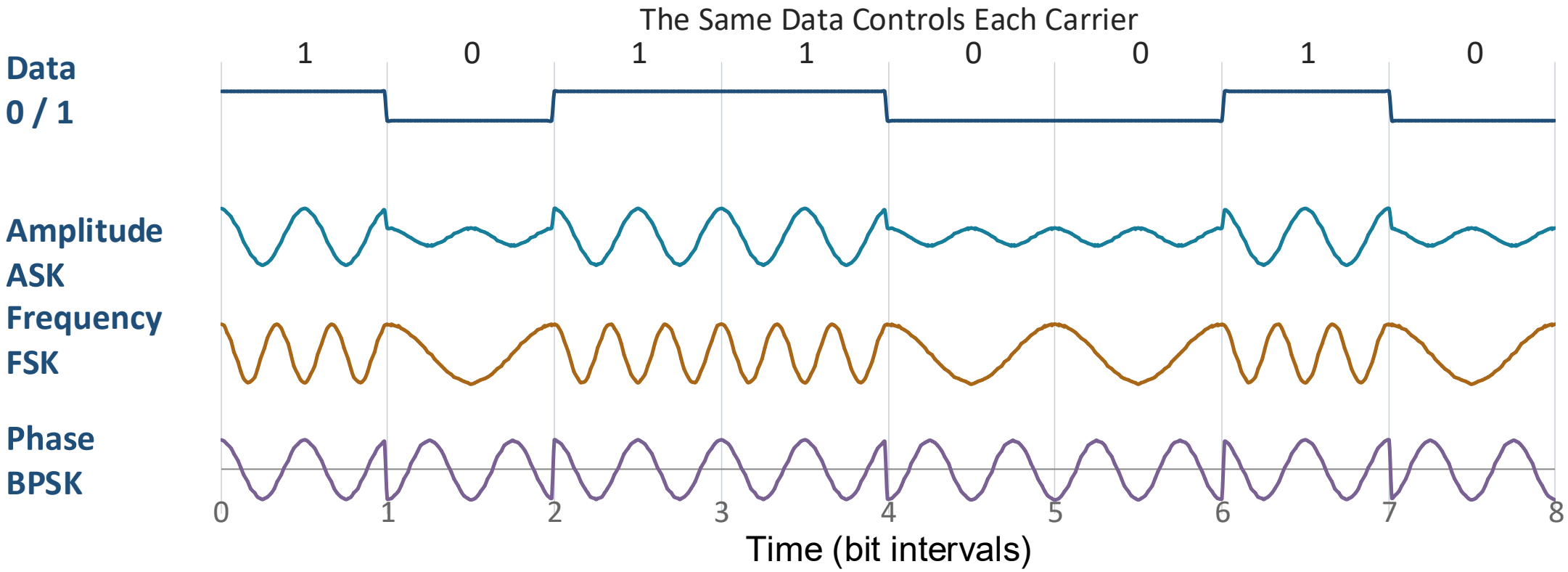


Four carrier phases represent four possible two-bit symbols.

Each transmitted symbol carries two bits.

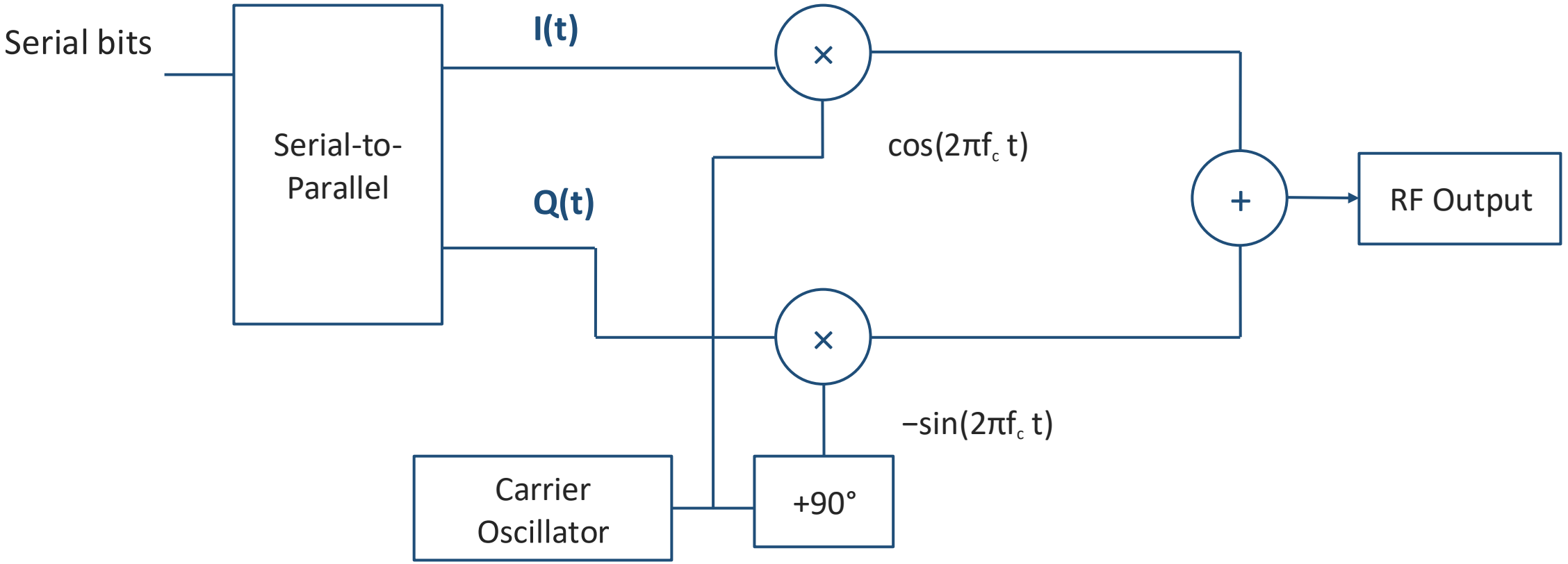
For an uncoded QPSK link, $R_s = R_b / 2$.

Summary: Amplitude, Frequency, and Phase



Vertical lines mark bit boundaries. The data waveform shows actual transitions; adjacent equal bits need no state change. Waveforms are illustrative.

The I and Q View of a Modulator



$$s(t) = I(t)\cos(2\pi f_c t) - Q(t)\sin(2\pi f_c t)$$

Question: BPSK or QPSK

Uncoded information rate = 1.0 Mbit/s. Keep received power, noise density, and pulse shaping fixed. Change BPSK to QPSK. Which statement is correct?

A. Symbol rate halves, so the RF bandwidth is about halved.

B. The received energy per information bit doubles.

C. The symbol rate doubles.

Worked Answer: BPSK or QPSK

A. Half the Symbol Rate and RF Bandwidth

$R_s = R_b$ / bits per symbol: BPSK = 1.0 Msymbol/s; QPSK = 0.5 Msymbol/s.

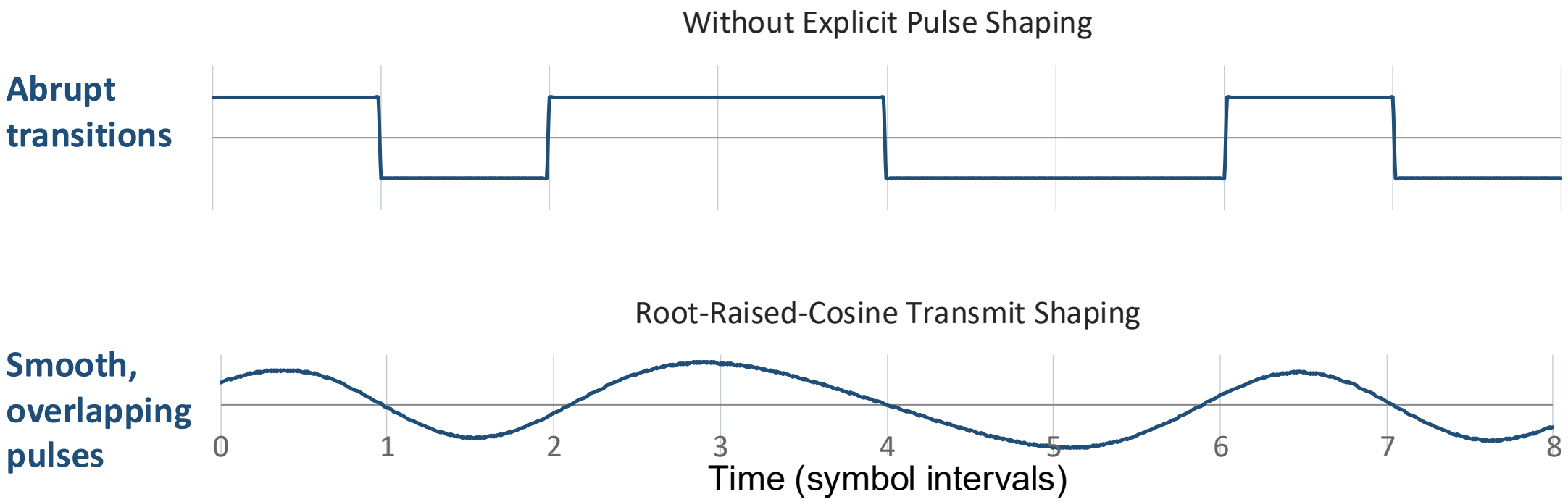
B is false: fixed power and information rate give the same energy per bit. C is false: symbol rate halves.

At a fixed SYMBOL rate instead, QPSK carries twice as many information bits per second.

Break

Next: Pulse Shaping, RF Bandwidth, and the Complete E_b/N_0 Budget

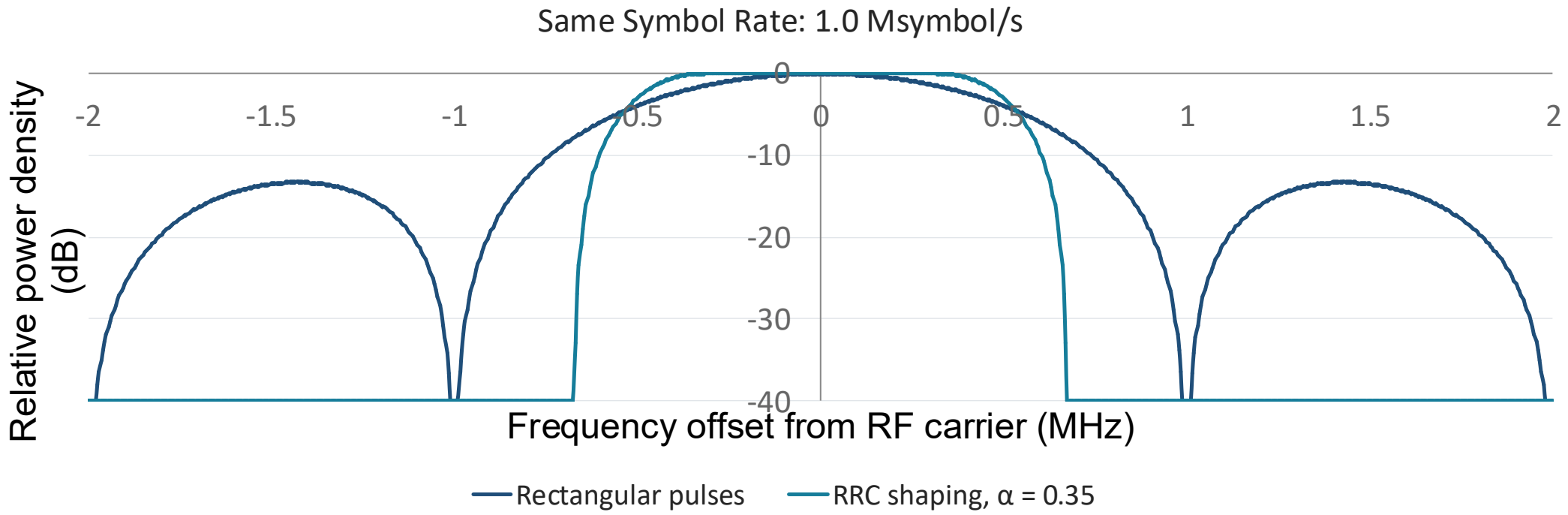
Pulse Shaping: The Same Bits in Time



Same eight bits and symbol rate. Only the transmit pulse shape changes.

Normalized baseband voltage, uncoded BPSK. RRC roll-off $\alpha = 0.35$. Pulse overlap is intentional; a matched receiver controls interference at decision instants.

Pulse Shaping: RF Bandwidth and Sidelobes



Rectangular: first-null span = $2R_s$ = 2.0 MHz; sidelobes extend beyond it.

Ideal RRC: RF span = $(1 + \alpha)R_s$ = 1.35 MHz. Practical filters need allowances.

Pulse Shaping: Benefits and Costs

Benefit	Cost or Constraint
Lower out-of-band emissions	Transmit and receive filters must work together
More controlled channel bandwidth	Filter memory and timing affect symbol decisions
Closer channel spacing	Conventional RRC-shaped BPSK* can need amplifier backoff
More efficient use of spectrum	Choose a waveform compatible with the spacecraft amplifier

*RRC = root-raised cosine. Saturation can distort envelope variations and regrow sidelobes.
GMSK (Gaussian minimum-shift keying) and selected shaped OQPSK (offset QPSK) can offer favorable amplifier behavior.

[13, 14]

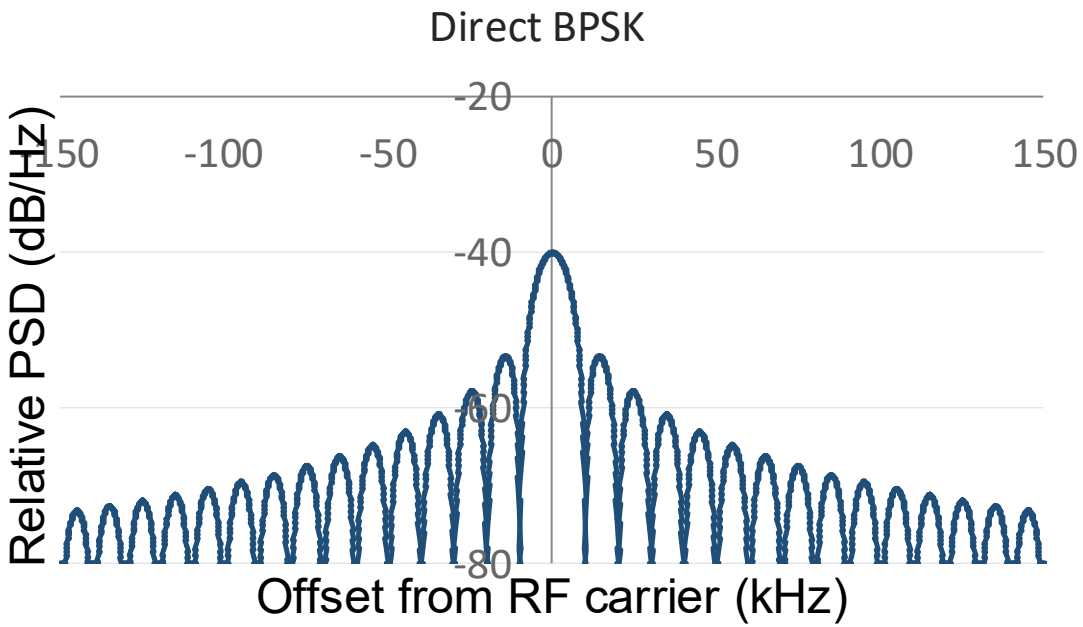
Bandwidth Example: BPSK and QPSK

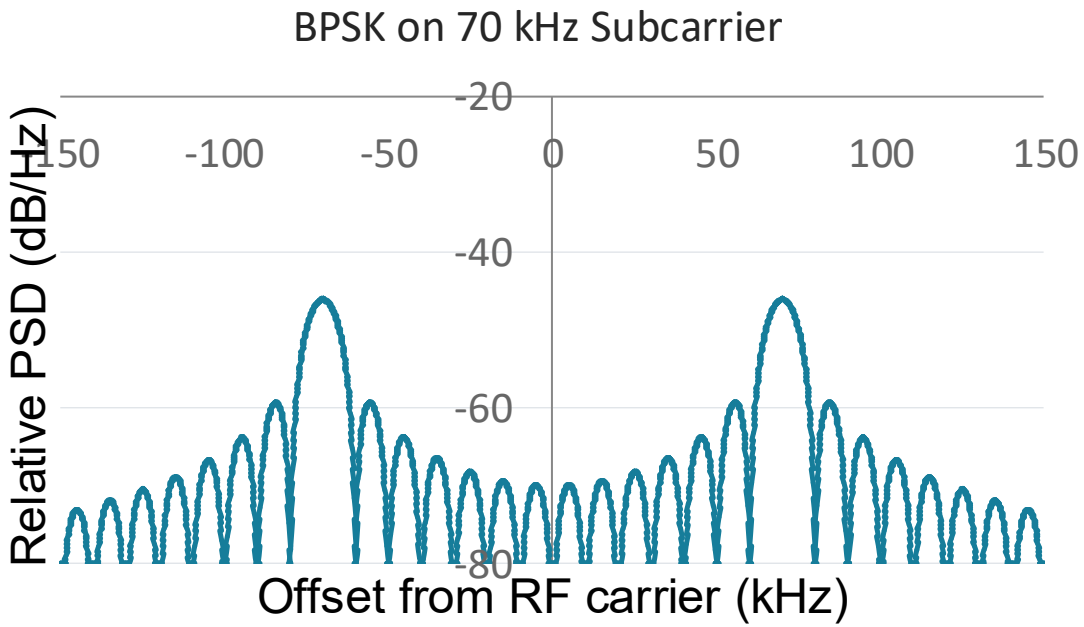
Assumptions	BPSK	QPSK
Information rate R_b	1.0 Mbit/s	1.0 Mbit/s
Bits per symbol	1	2
Symbol rate R_s	1.0 Msymbol/s	0.5 Msymbol/s
B_{RF} at $\alpha = 0.35$	1.35 MHz	0.68 MHz

Uncoded examples. $B_{RF} = (1 + \alpha)R_s$. Both transmit the same number of information bits each second.

[3]

Direct Modulation and Subcarriers





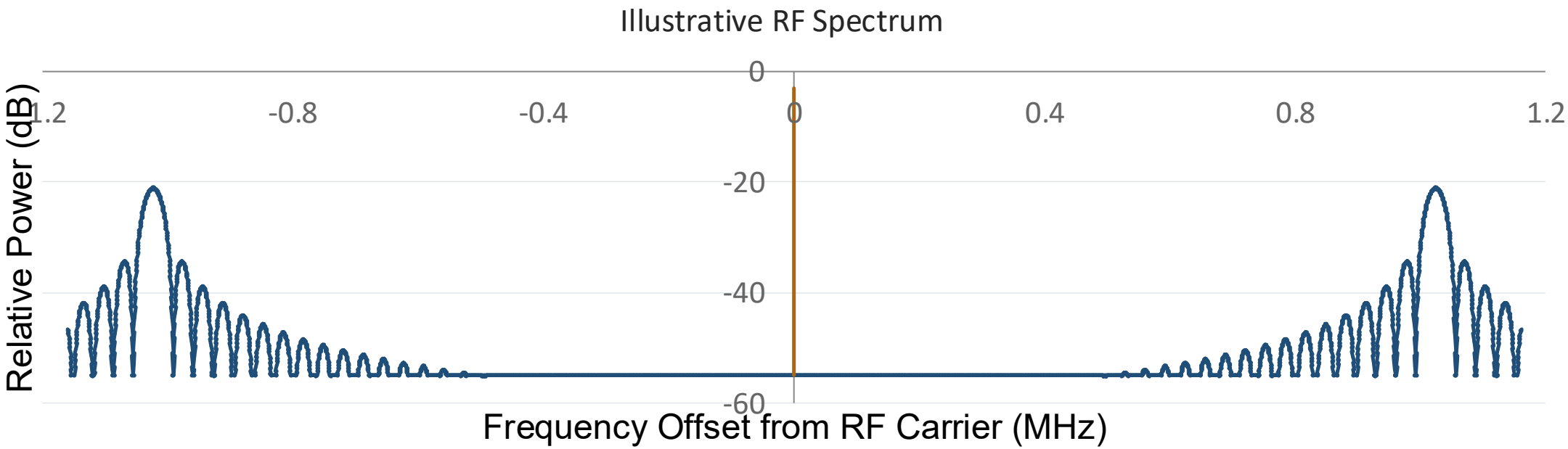
Data spectrum is centered on the RF carrier.

Data sidebands move to ±70 kHz.

*Calculated average spectra: uncoded 10 kbit/s NRZ, no pulse shaping. PSD relative to unit complex-envelope power/Hz. No receiver noise; display stops at −80 dB/Hz.

Residual Carrier With a BPSK Subcarrier

Lower Sideband **Residual Carrier** **Upper Sideband**



−1.024 MHz **0** **+1.024 MHz**

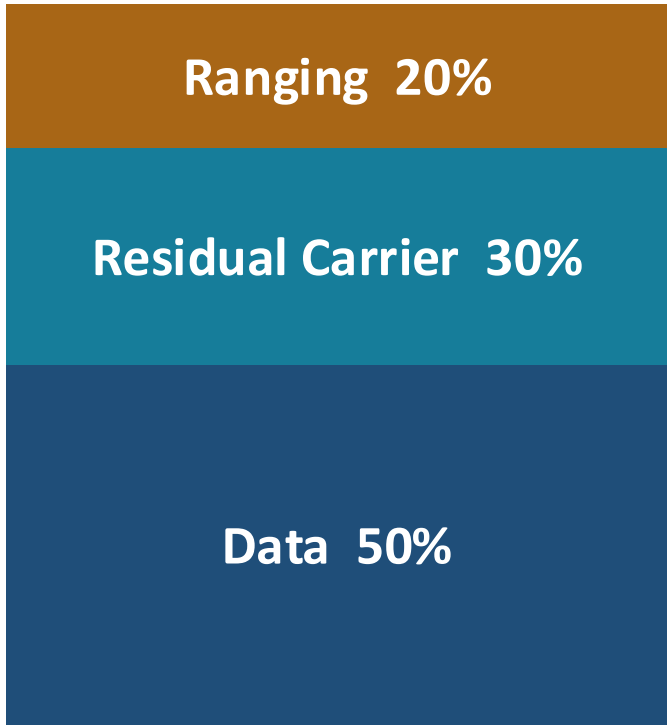
32 kbit/s BPSK on a 1.024 MHz subcarrier

*Illustrative power allocation. Unshaped NRZ data: each sideband has a 64 kHz first-null span.

Data Power Is Part of Total Signal Power

Illustrative Allocation of Total Received RF Power

100% of Total RF Power



$$q = S_{data} / S_{total} = 0.5$$

$$[S_{data}/N_0] = [S_{total}/N_0] + 10 \log_{10}(q)$$

For q = 0.5, data-channel S/N_0 is 3.0 dB lower.

*Assumed power fractions sum to 100%. Modulation indices and waveform design determine the actual split. The diagram shows allocation, not signal flow.

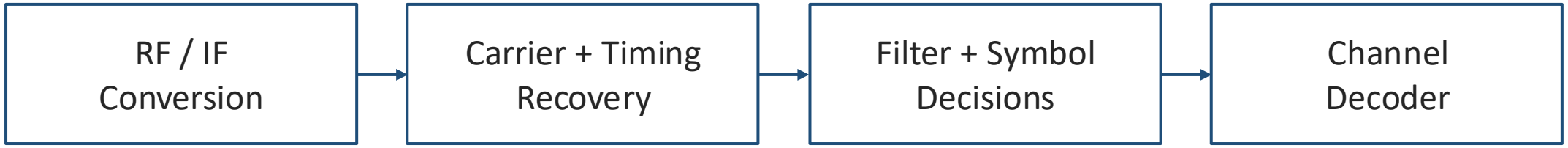
Modulation Selection Is a Mission Trade

Need	Signal Design Implication
Weak signal acquisition	Consider tracking aids and residual-carrier power
High data rate in limited spectrum	More bits per symbol and appropriate pulse shaping
Efficient spacecraft power amplifier	Control envelope variation and amplifier backoff
Navigation or tracking on the same link	Allocate signal power and spectrum to each function

Choose the waveform with the receiver, amplifier, and mission requirements in view.

[3, 4, 6]

The Receiver Must Recover the Bits



The receiver needs carrier and timing references before making symbol decisions.

Its error performance depends on signal energy relative to noise.

Next class examines the receiver and decoder in detail.

Bit Errors and Mission Performance

1,000,000 Bits

BER = 10^{-5}

10 Bit Errors on Average

Measured BER = erroneous bits / tested bits

A theoretical bit-error probability predicts long-run behavior under stated assumptions.

At BER = 10^{-5}, one million bits contains 10 errors on average.

A mission may require frame or packet error performance, not only raw BER.

Energy per Information Bit

Energy per Bit = Received Data Power ÷ Information Bit Rate

$E_b = S_{data} / R_b$

J/bit = (J/s) ÷ (bit/s)

$E_b/N_0 = (S_{data}/N_0) / R_b$

$[E_b/N_0] = [S_{data}/N_0] - 10 \log_{10}(R_b / 1 \text{ bit/s})$

At fixed data power, twice as many bits each second means half the energy per bit: −3.0 dB.

Lunar Link: Available E_b/N_0

Carry forward $[S_{data}/N_0]$ = 74.5 dB-Hz. Assume all signal power carries data.

At R_b = 1.0 Mbit/s:

$[E_b/N_0] = 74.5 - 10 \log_{10}(1{,}000{,}000) = 14.5$ dB

At R_b = 2.0 Mbit/s: 74.5 − 63.0 = 11.5 dB

Question: Doubling the Information Rate

Keep received data power and noise density fixed. Double information rate from 1.0 to 2.0 Mbit/s. What changes?

A. E_b/N_0 improves by 3.0 dB

B. E_b/N_0 worsens by 3.0 dB

C. S/N_0 worsens by 3.0 dB

Worked Answer: Doubling the Information Rate

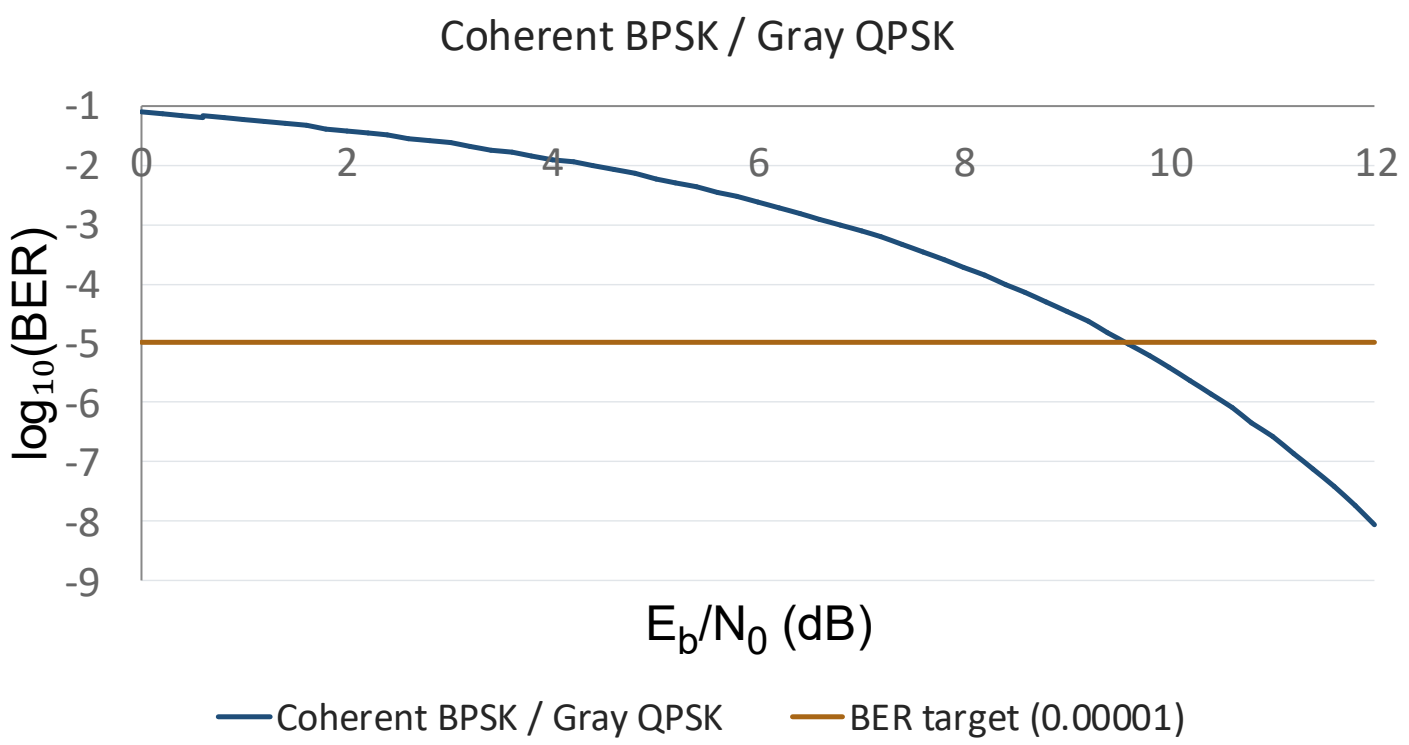
B. Available E_b/N_0 Falls by 3.0 dB

$[E_b/N_0] = [S_{data}/N_0] - [R_b]$. Rate rises from 60.0 to 63.0 dB(bit/s).

S_{data}/N_0 stays 74.5 dB-Hz. Available E_b/N_0 falls from 14.5 to 11.5 dB.

A has the sign reversed. C is false because the received power and noise density stay fixed.

The Ideal BPSK and QPSK BER Curve



−5 on the vertical axis means BER = 10^{-5}.

Coherent BPSK and Gray QPSK in AWGN:

$P_b = \frac{1}{2}\,\mathrm{erfc}(\sqrt{(E_b/N_0)})$

E_b/N_0 inside the square root is a linear ratio.

At BER = 10^{-5}, the ideal requirement is approximately 9.6 dB.

Reading a BER Curve

E_b/N_0	Ideal Uncoded BER	Interpretation
6.0 dB	2.4×10^{-3}	About 2,400 errors per million bits
9.6 dB	9.7×10^{-6}	About 10 errors per million bits
12.0 dB	9.0×10^{-9}	Far fewer errors under this ideal model

Coherent BPSK, AWGN, perfect synchronization. Actual equipment adds implementation effects.

[7]

Available and Required E_b/N_0

Available E_b/N_0 comes from received data power, noise density, and information rate.

Required E_b/N_0 comes from a performance curve or a tested receiver specification.

Example: ideal coherent uncoded BPSK needs about 9.6 dB for BER = 10^{-5} in AWGN.

Real receiver losses and design reserve must also fit in the budget.

Lunar Link: Closing the Margin

Available $[E_b/N_0] = [S_{data}/N_0] - [R_b] = 74.5 - 60.0 = 14.5$ dB

Assume BER $\leq 10^{-5}$ requires 9.6 dB for ideal uncoded BPSK.

Assume a separate implementation loss of 1.0 dB.

Margin = 14.5 − 9.6 − 1.0 = 3.9 dB

Lunar Link: Rate and Margin

Information Rate	Available E_b/N_0	Margin*
1.0 Mbit/s	14.5 dB	3.9 dB
2.0 Mbit/s	11.5 dB	0.9 dB
4.0 Mbit/s	8.5 dB	−2.1 dB

*Margin = $74.5 - 10\log_{10}(R_b) - 9.6 - 1.0$ dB. Bandwidth and lock must also be adequate.

Question: The 4 Mbit/s Link

We need 3.0 dB reserve, but the 4.0 Mbit/s case has −2.1 dB margin. Which change alone is enough?

A. Switch BPSK to QPSK at the same information rate

B. Reduce the information rate to 1.0 Mbit/s

C. Add 20 dB of ideal gain after the receiver front end

Worked Answer: The 4 Mbit/s Link

B. Reduce the Information Rate

The 1.0 Mbit/s case has 3.9 dB margin and exceeds the 3.0 dB reserve.

A reduces bandwidth but does not change ideal bit-energy performance.

C amplifies signal and existing noise together.

Lunar Link: The Remaining Trades

Change	Budget Effect	Mission Cost
More data-channel EIRP	More available E_b/N_0	Spacecraft power or antenna demands
Better G/T or lower losses	More available E_b/N_0	Ground hardware, siting, or operations
Lower information bit rate	More energy per information bit	Longer delivery time
Different modulation / coding	May lower required E_b/N_0	Bandwidth, complexity, or latency

A lower requirement must come from a justified performance comparison at the same delivered error target.

[3, 5, 7]

The Complete Link Equation

$$[E_b/N_0] = [EIRP] - [L_p] - [L_{other}] + [G/T] - [k] + [q] - [R_b]$$

$[q] = 10 \log_{10}(q)$, where q is the fraction of received signal power allocated to data.

$[R_b] = 10 \log_{10}(R_b / 1\ \text{bit/s})$, using the information bit rate.

$[k] = -228.6$ dBW/Hz/K. G and T_{sys} use the same reference plane.

Available E_b/N_0 must cover the requirement, implementation loss, and reserve.

What If the Link Still Does Not Close?

We have optimized power, antennas, receiver noise, losses, and data rate.

Our available E_b/N_0 still falls short of the requirement.

Could we transmit the bits in a way that requires less E_b/N_0?

Next Class: Demodulation, BER, and Coding

Optional Reference Slides

Additional Signal Examples and Calculation Details

Why the 290 K Reference Cancels

Chosen Reference T_0	Reported Noise Factor F	Recovered Receiver T_e
290 K	2.0 (NF = 3.0 dB)	(2.0 − 1) × 290 = 290 K
Hypothetical 580 K	1.5 (NF = 1.8 dB)	(1.5 − 1) × 580 = 290 K

$T_e = (F - 1)T_0$. Same hardware, different reporting convention. Actual output noise = $GkB(T_A + T_e)$, unchanged by the reference choice.

[2]

Pulse-Shape Mismatch: A Numerical Example

Receive Filter	Required E_b/N_0 at BER = 10^{-5}
Matched root-raised-cosine filter	9.6 dB
Rectangular integrate-and-dump filter	10.5 dB
Additional required E_b/N_0	About 1.0 dB

*Teaching calculation: uncoded BPSK, RRC transmit roll-off 0.35, AWGN, perfect carrier/timing, independent bits, no equalizer. This penalty is specific to these assumptions.

[3, 13]

Navigation and Tracking Signals

Purpose	Observable or Reference	Example
Navigation	Known timing/code/phase references	GPS time-tagged ranging codes
Historical navigation	Carrier Doppler over a pass	Transit navigation
Bearing reference	Carrier/subcarrier phase relationship	VOR
Spacecraft tracking	Range and Doppler over time	Orbit estimation

Range is a line-of-sight distance measurement. Doppler constrains line-of-sight velocity; a trajectory solution uses measurements over time.

[4]

Frequency Bands and Spectrum Access

S, X, Ku, and Ka are useful space-communications bands. L, VHF, and UHF also serve particular missions.

Available bandwidth depends on the allocated service, direction, region, and authorization.

Higher-frequency links can offer wider allocations while facing tighter pointing and weather trades.

A band name alone does not guarantee a usable bandwidth or non-line-of-sight coverage.

Subcarrier Examples and Ranging Tones

Direct example: 100 kbit/s telemetry modulates the RF carrier without an added subcarrier.

Subcarrier example: place 100 kbit/s data on a 1.024 MHz subcarrier. Account for both RF sidebands.

Combined example: 20 kbit/s telemetry on a 70 kHz subcarrier, plus ranging tones at 4, 20, 100, and 500 kHz.

These are illustrative plans. Modulation indices set the power split; receiver compatibility sets usable combinations.

PN ranging uses a known pseudo-random pattern and correlation. It need not use a telemetry subcarrier.

Analog Modulation Equations

AM: $s(t) = A[1 + \mu m(t)] \cos(2\pi f_c t)$

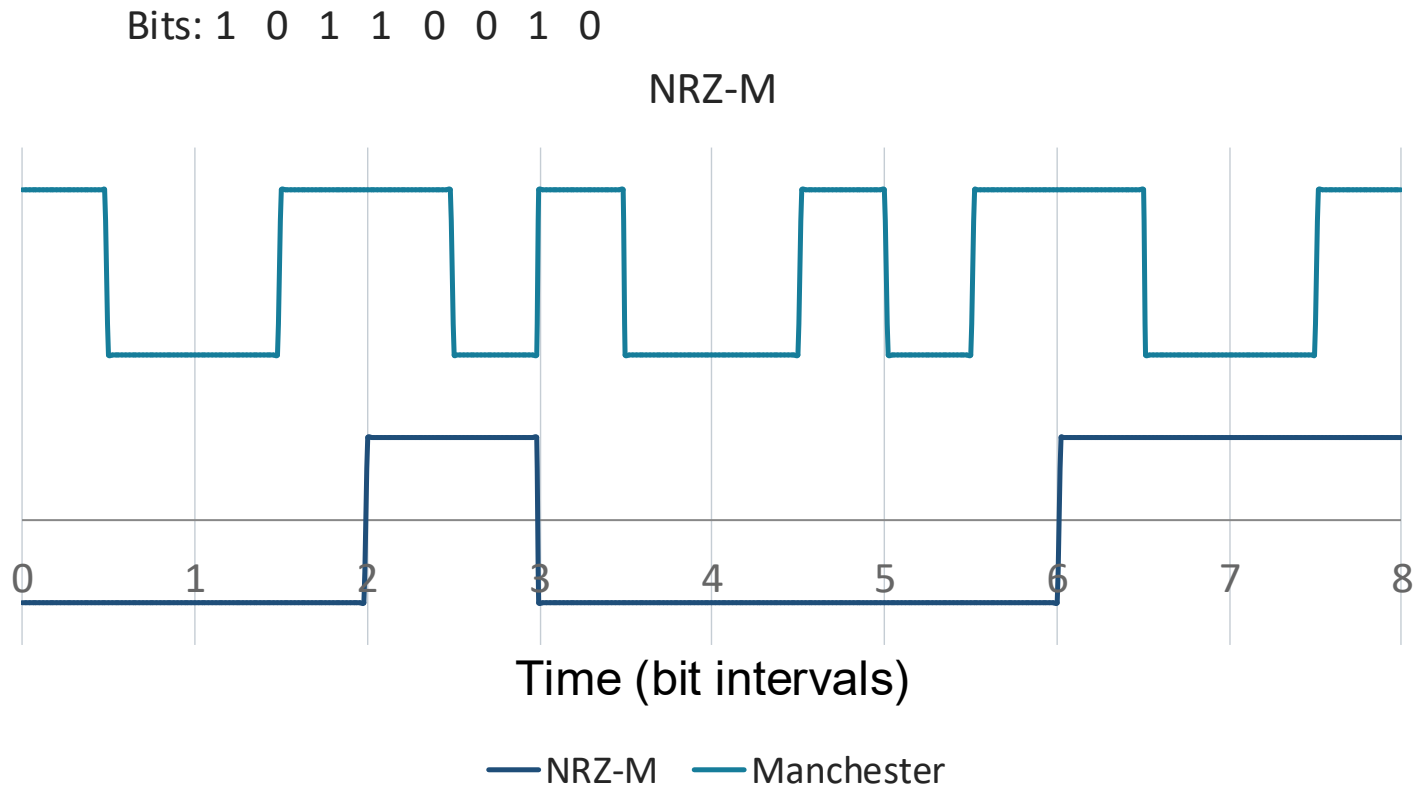
FM: $s(t) = A \cos[2\pi f_c t + 2\pi k_f \int m(\tau)d\tau]$

PM: $s(t) = A \cos[2\pi f_c t + k_p m(t)]$

For FM, Carson's approximation is $B \approx 2(\Delta f + f_m,max)$.

No blanket AM/FM/PM performance ranking applies without assumptions.

Manchester, NRZ, and Timing Recovery



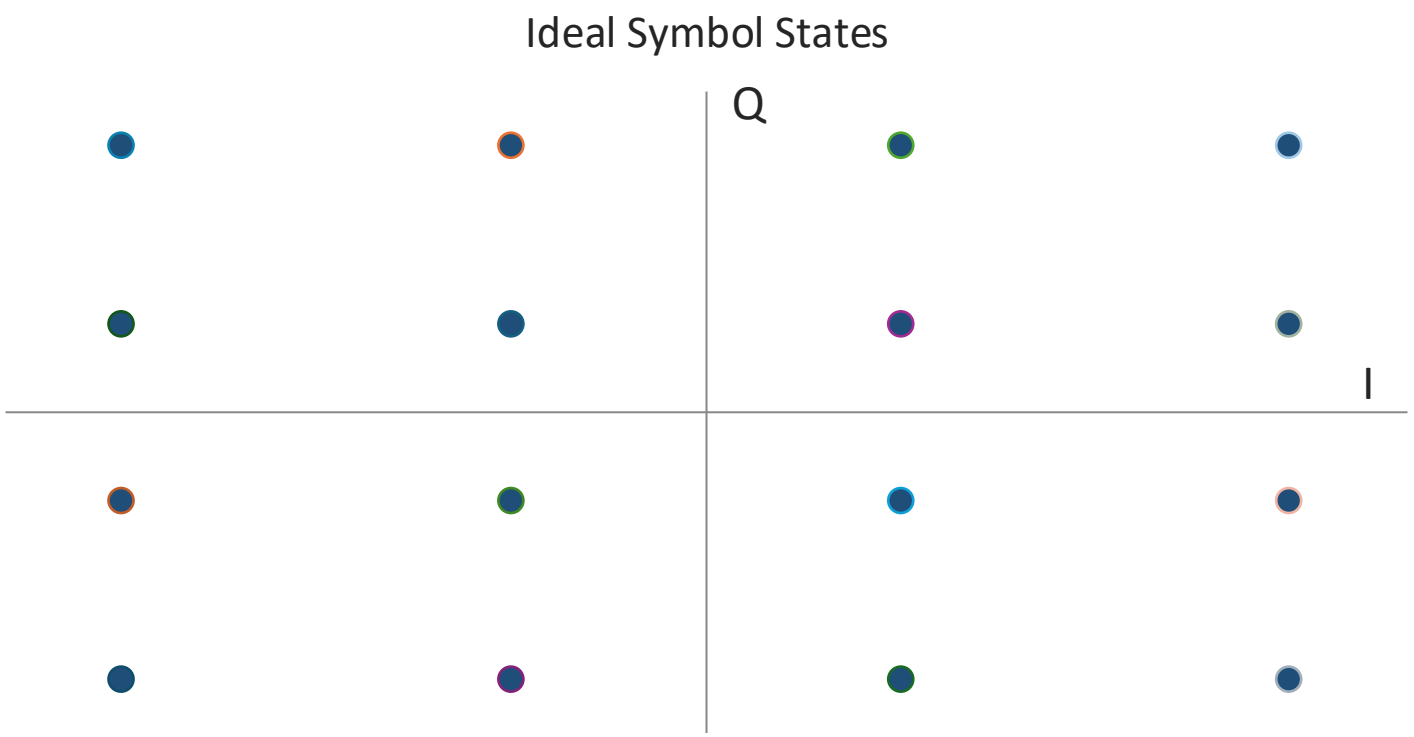
Long runs without transitions make timing recovery harder.

Manchester guarantees a transition at each bit center.

NRZ schemes can use scrambling or other structure to aid timing.

A line-code symbol, a coded bit, and an RF modulation symbol are different concepts.

Higher-Order Constellations



M-PSK places more phase states on a circle. QAM uses amplitude and phase.

More coded bits per symbol can reduce the required symbol rate.

At fixed average symbol energy, more closely spaced states are harder to distinguish.

Envelope variation and amplifier linearity become important.

Lunar Link: Maximum Rate With Reserve

Require 3.0 dB design reserve beyond the BER requirement and implementation loss.

$[R_b,\max] = 74.5 - 9.6 - 1.0 - 3.0 = 60.9$ dB(bit/s)

$R_b,\max = 10^{60.9/10} \approx 1.2$ Mbit/s

This is a power-limited result. Spectrum, coding, hardware, and acquisition can lower it.

Lunar Link: Reference Planes

Quantity	Antenna Terminals A	LNA Input B
Received signal	−131.9 dBW	−132.9 dBW
Equivalent system temperature	168.0 K / 22.3 dBK	133.5 K / 21.3 dBK
Equivalent noise density	−206.3 dBW/Hz	−207.3 dBW/Hz
S/N_0	74.5 dB-Hz	74.5 dB-Hz

Same complete system represented at two planes. B includes receiver-added noise referred to its input. Inserting a lossy cable physically still degrades the link.

References and Further Reading

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[2] Keysight. Noise Figure application help: thermal noise and the 290 K reference.

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[4] JPL. DSN Telecommunications Link Design Handbook, 810-005, Module 206: Telemetry General Information.

[5] CCSDS 131.0-B-5 (2023). TM Synchronization and Channel Coding.

References and Further Reading

[6] Keysight. Digital Modulation in Communications Systems, Application Note 1298.

[7] MathWorks. Analytical Expressions Used in berawgn Function and Bit Error Rate Analysis App.

[8] CCSDS 130.1-G-3 (2020), with technical corrigendum: TM Synchronization and Channel Coding, Summary of Concept and Rationale.

[9] MIT 6.450, Lecture 23: channel capacity and the wideband energy limit.

[10] NTIA. United States Frequency Allocation Chart and Table of Frequency Allocations.

References and Image Credits

[11] NASA/Goddard Space Flight Center. LRO Assembly and Testing Stills, SVS 10408. Spacecraft photograph.

[12] NASA/JPL-Caltech. Goldstone 34-meter Beam Waveguide, PIA17791. Ground antenna photograph.

[13] MathWorks. Raised Cosine Filtering and Raised Cosine Receive Filter documentation.

[14] CCSDS 413.0-G-3, with Technical Corrigendum 1. Bandwidth-Efficient Modulations, Sections 3 and Annex B.

[15] Spacecraft–Earth scene: AI-generated conceptual illustration. Distances and sizes are not to scale.